



CSE 641 Computer Vision

Weekly Report 5

Automated Segmentation of Post-Treatment Glioma Sub-regions and Resection Cavities from Multi-Parametric MRI.

Submitted to faculty: Prof. Mehul Raval

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Student Details

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Aim:

To develop and evaluate deep learning architectures for the accurate, automated 3D segmentation of brain tumors using the BraTS dataset, translating raw MRI scans into clinically actionable metrics.

Introduction:

Accurate segmentation of gliomas is very important for neurosurgery planning. The BraTS dataset contains 3D MRI scans from different angles, which pose a significant computational challenge due to its large size and the diverse shapes of tumors.

This week, we focused on building a 3D Convolutional Neural Network (CNN) from scratch, training it, and testing it to evaluate its performance. The goal is to lay a foundation for future work on gliomas and 3D CNN and train the SegResNet model.

Work Completed:

Data Pipeline Construction: We successfully loaded the Brats dataset and split it into 80% training and 20% internal validation using MONAI. We also implemented `randspatcropd` to extract 96x96x96 3D patches, which helps with efficient GPU training without memory exhaustion.

- **Baseline Model Training:** We Configured and trained a SegResNet model from scratch for 50 epochs using PyTorch, utilizing Mixed Precision to speed up training times.
- **Advanced Evaluation Metrics:** Built an evaluation script to calculate both the Dice Similarity Coefficient (DSC) for volume overlap and the 95% Hausdorff Distance (HD95) for boundary accuracy.
- **Clinical Translation:** Developed an automated script that translates pixel predictions into real-world volumetric data (Cubic Centimeters) for specific tumor sub-regions.
- **Visualization Pipeline:** We also wrote a plotting script to dynamically find the MRI slice with the largest tumor and display the T1/FLAIR scan, the Doctor's Ground Truth, and the Prediction side-by-side for qualitative review.

Results:

The SegResNet baseline successfully learned human brain anatomy and tumor feature extraction, achieving the following results on the internal validation set:

- **Mean Lesion-wise DSC: 0.7714** (A very strong initial baseline, proving the model is accurately capturing the bulk of the tumor volume).

- **Mean Lesion-wise HD95: 16.0187 mm** (Indicates occasional model "hallucinations" of distant tumor pixels, which slightly inflate the boundary error penalty).

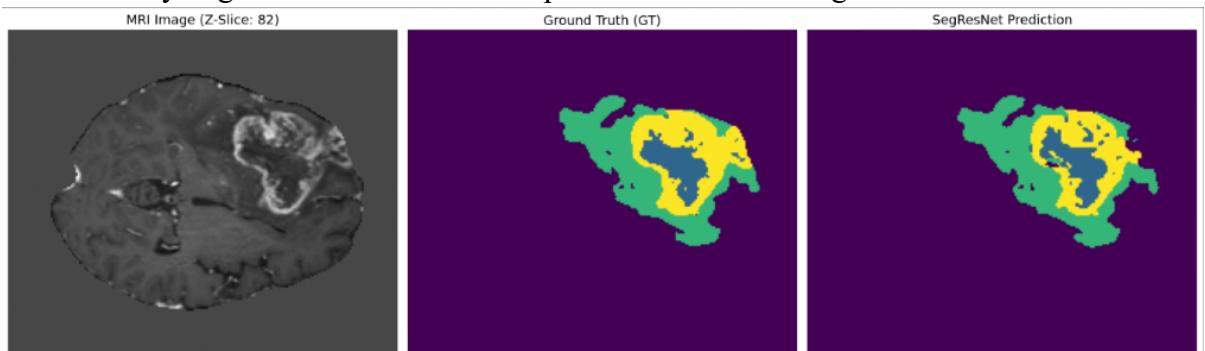
```
Advanced Evaluation Complete!
Mean Lesion-wise Dice Score (DSC): 0.7714
Mean Lesion-wise 95% Hausdorff Distance (HD95): 16.0187 mm
```

- **Clinical Volume Example:** Extracted real-world metrics for a sample patient (Tumor Core: **65.13 cm³**, Surrounding Edema: **174.54 cm³**).

```
Generating Clinical Volume Report for a Patient...

=====
                        CLINICAL VOLUME REPORT
=====
Tumor Core (ET + NETC) Volume : 65.13 cm3
Surrounding Edema (SNFH) Vol  : 174.54 cm3
=====
```

- **Visual Output:** The generated visualization plots confirm that the predicted tumor masks closely align with the structural shape and location of the ground truth masks.



Next steps and goals:

Train Swin UNETR: Set up and begin training the Swin UNETR model (a Vision Transformer). Transformers use global self-attention rather than localized sliding windows, which should theoretically improve segmentation accuracy.

- **Apply Post-Processing:** Implement a "Keep Largest Connected Component" algorithm on the SegResNet predictions to remove isolated background noise and drastically lower the 16 mm HD95 error.
- **Model Comparison:** Create a comparative analysis script to evaluate SegResNet (CNN) vs. Swin UNETR (Transformer) across DSC, HD95, and inference speeds.

Conclusion:

This week, we completed the end-to-end pipeline, from data loading to clinical evaluation and visualization. It is now fully functional. The SegResNet baseline achieved a highly

respectable 77% Dice Score, proving the data preprocessing and training loops are working perfectly. We are now well-positioned to move on to more advanced Transformer-based architectures next week.

References:

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2. M. C. de Verdier et al., “The 2024 Brain Tumor Segmentation(BraTS) Challenge: Glioma Segmentation on Post-treatment MRI,” arXiv preprint arXiv:2405.18368, 2024.
3. A. Parida et al., “Improving Pre-trained Adult Glioma Segmentation Models using only Post-processing Techniques,” arXiv preprint arXiv:2512.14937, 2025.
4. H. Kim et al., “Effective Segmentation of Post-Treatment Gliomas Using Simple Approaches: Artificial Sequence Generation and Ensemble Models,” arXiv preprint arXiv:2409.08143, 2024.
5. A. Ferreira et al., “Improved Multi-Task Brain Tumour Segmentation with Synthetic Data Augmentation,” arXiv preprint arXiv:2411.04632, 2024.
6. A. Celaya et al., “Pre- and Post-Treatment Glioma Segmentation with the Medical Imaging Segmentation Toolkit,” arXiv preprint arXiv:2507.19626, 2025.